

Online Shopping Sales

Trends & Insights



Problem Statement

The online shopping business is facing challenges in identifying sales trends, which are essential for management to understand key factors influencing revenue. There is a need for deeper insights into product performance, customer purchasing behavior, and sales trends over time. Our aim is to uncover patterns in revenue, customer demographics, and the impact of pricing, enabling data-driven decisions to optimize marketing strategies and enhance overall business performance.

Dataset Source

The dataset is sourced from Kaggle and it contains online shopping transaction records.

<https://www.kaggle.com/datasets/jacksondivakarr/online-shopping-dataset>

Data Preparation Steps

- ❑ Removed 31 rows (0.058%) that had missing values across multiple columns to ensure data consistency.
- ❑ Deleted irrelevant columns that did not contribute to the analysis.
- ❑ Verified and confirmed that there were no duplicate records.
- ❑ Created a new column for revenue by multiplying the unit price of a product by its quantity.
- ❑ Standardized data types for consistency across text, numeric, currency, and date fields.
- ❑ Created bins for unit price and tenure month to segment the data for analysis.

Key Insights from the Dashboard

- ❑ Female customers contribute to a larger share of total revenue (62.17%) compared to male customers (37.83%).
- ❑ Revenue fluctuate across the year, with a noticeable increase towards the end of the year, indicating possible seasonal demand.
- ❑ Products belong to office generate the highest sales volume, followed by Apparel and Drinkware.
- ❑ Some cities, such as Chicago and California outperform others in revenue, suggesting opportunities for targeted marketing and expansion in high-performing regions.
- ❑ A large proportion of purchases are made at lower price points, indicating price sensitivity among customers.
- ❑ Customer retention is strong up to 36 months but declines sharply after that , indicating a lower long-term customer association with the business.

Thank You